

Spring Data JPA - Quick Example

OrmLearnApplication.java:

```
package com.cognizant.orm_learn;

import com.cognizant.orm_learn.model.Country;
import com.cognizant.orm_learn.service.CountryService;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
import java.util.List;

@SpringBootApplication
public class OrmLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(OrmLearnApplication.class);
    private static CountryService countryService;

    public static void main(String[] args) {

        ApplicationContext context =
            SpringApplication.run(OrmLearnApplication.class, args);

        countryService = context.getBean(CountryService.class);

        LOGGER.info("Inside main");

        testGetAllCountries();

    }

    private static void testGetAllCountries() {

        LOGGER.info("Start");

        List<Country> countries = countryService.getfullTable();

        LOGGER.debug("Countries = {}", countries);

        LOGGER.info("End");

    }

}
```

OUTPUT:

```
Run OrmLearnApplication x
Console Health Mappings Environment
05-07-26 11:48:00 INFO com.cognizant.orm_learn.OrmLearnApplication - Started OrmLearnApplication seconds (process running for 4.04
05-07-26 11:48:00 INFO com.cognizant.orm_learn.OrmLearnApplication - Inside main
05-07-26 11:48:00 INFO com.cognizant.orm_learn.OrmLearnApplication - Start
05-07-26 11:48:00 DEBUG org.hibernate.SQL - select c1_0.co_code, c1_0.co_nm from country c1_0
05-07-26 11:48:00 INFO com.cognizant.orm_learn.OrmLearnApplication - Countries = [Country--Table:
CodeIN
Name:INDIA, Country--Table:
CodeUS
Name:United States of America]
05-07-26 11:48:00 INFO com.cognizant.orm_learn.OrmLearnApplication - End
```

Difference between JPA, Hibernate and Spring Data JPA

Feature	JPA	Hibernate	Spring Data JPA
Full Form	Java Persistence API	Hibernate ORM	Spring Data JPA
Type	Specification (API)	ORM Framework	Spring Module
Developed By	Oracle (Java)	Hibernate Team	Spring Team
Purpose	Defines rules for ORM	Implements JPA and provides ORM features	Simplifies JPA by reducing boilerplate code
SQL Writing	Uses JPQL	Uses HQL/JPQL	Automatically generates queries from method names
Dependency	Requires an implementation	Can work as a JPA implementation	Uses JPA internally

